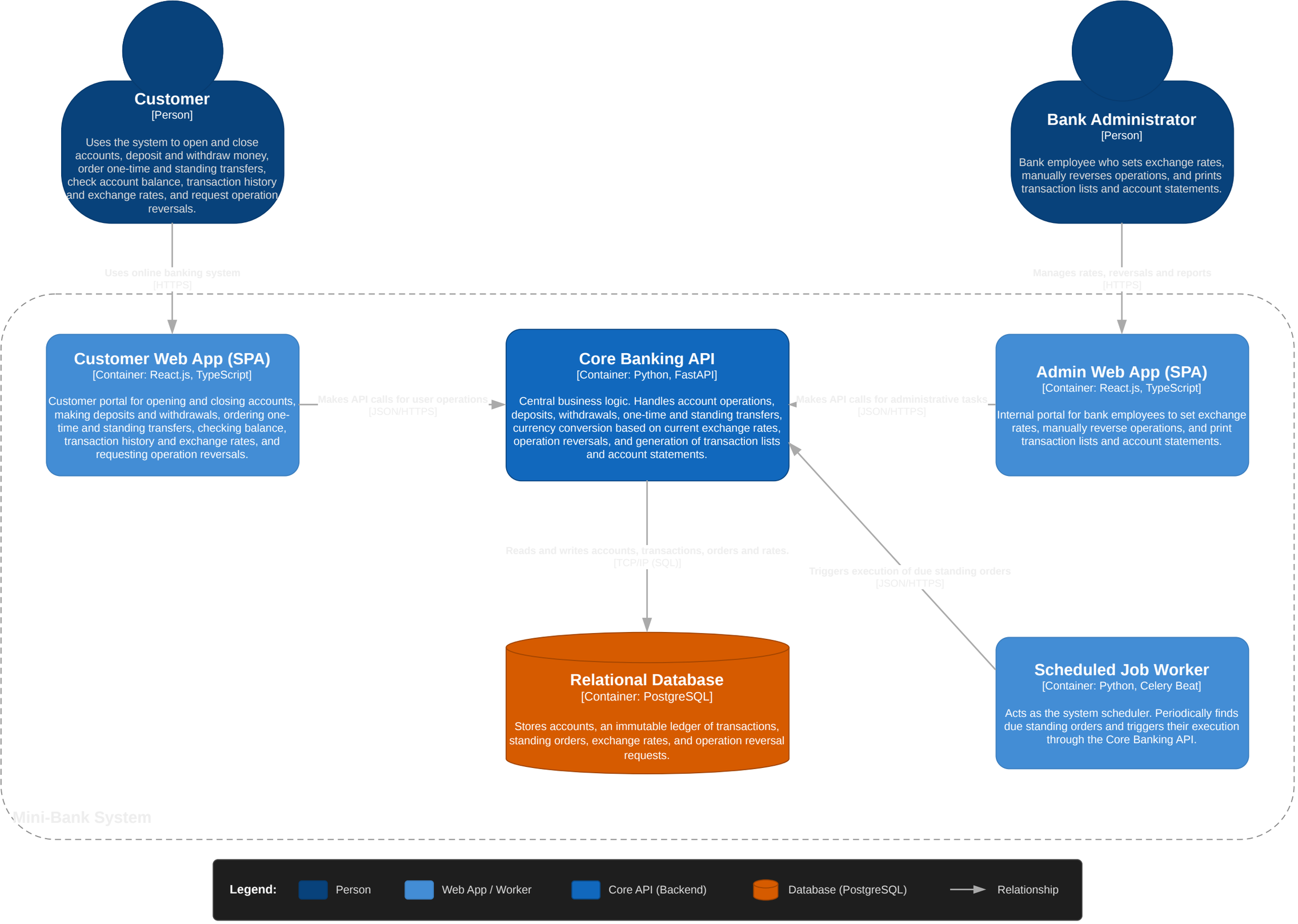
Customer
[Person]
Uses the system to open and close accounts, deposit and withdraw money, order one-time and standing transfers, check account balance, transaction history and exchange rates, and request operation reversals.
Bank Administrator
[Person]
Bank employee who sets exchange rates, manually reverses operations, and prints transaction lists and account statements.
Uses online banking system
[HTTPS]
Manages rates, reversals and reports
[HTTPS]
Customer Web App (SPA)
[Container: React.js, TypeScript]
Customer portal for opening and closing accounts, making deposits and withdrawals, ordering one-time and standing transfers, checking balance, transaction history and exchange rates, and requesting operation reversals.
Makes API calls for user operations
[JSON/HTTPS]
Core Banking API
[Container: Python, FastAPI]
Central business logic. Handles account operations, deposits, withdrawals, one-time and standing transfers, currency conversion based on current exchange rates, operation reversals, and generation of transaction lists and account statements.
Makes API calls for administrative tasks
[JSON/HTTPS]
Admin Web App (SPA)
[Container: React.js, TypeScript]
Internal portal for bank employees to set exchange rates, manually reverse operations, and print transaction lists and account statements.
Reads and writes accounts, transactions, orders and rates.
[TCP/IP (SQL)]
Triggers execution of due standing orders
[JSON/HTTPS]
Relational Database
[Container: PostgreSQL]
Stores accounts, an immutable ledger of transactions, standing orders, exchange rates, and operation reversal requests.
Scheduled Job Worker
[Container: Python, Celery Beat]
Acts as the system scheduler. Periodically finds due standing orders and triggers their execution through the Core Banking API.
Mini-Bank System
Legend:
Person
Web App / Worker
Core API (Backend)
Database (PostgreSQL)
Relationship